Qn. Set Code-7

Semester: 4th
Programme: B.Tech
Branch: CSE, IT, CSCE, CSSE, E&CSc

SPRING END SEMESTER EXAMINATION-2025

4th Semester B.Tech

DISCRETE MATHEMATICS

MA21002

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

   (a) Find the inverse and contrapositive of the statement "I go to college whenever my friends come".

   (b) Translate the English sentence into predicate logic "some students in this class has taken a course in Java", where the domain consists of all people.

   (c) Find the number of reflexive and symmetric relations on a set with 6distinct elements.

   (d) Find the congruence class modulo$m$ of a nonzero integer k.

   (e) How many positive integers not exceeding 1000 are divisible by 5 or 7?

   (f) Find the general solution of the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$, for $n \geq 2$.

(g) Is the group $G = \{1, w, w^2\}$ under complex multiplication cyclic? If yes, find its generators.

(h) Define complete graph. If a complete graph has degree 5 for each vertex, then find the number of edges in the graph.

(i) Draw the Hasse diagram of the POSET $(D_{20}, |)$, where D_{20} is the set of all positive divisor of 20.

(j) Define zero divisors in a ring. Find the zero divisors in the ring$(\mathbb{Z}_6, \oplus_6, \otimes_6)$.

SECTION-B

2. (a) Use strong mathematical induction to prove that "Every integer $n \geq 2$ is a product of primes." [5]

(b) If $(R, +, \cdot)$ is a ring then prove that [5]

(i) $a \cdot 0 = 0, \forall a \in R$

(ii) $a \cdot (-b) = -(a \cdot b), \forall a, b \in R$

(iii) $(-a) \cdot (-b) = a \cdot b, a, b \in R$

3. (a) Show that $(p \to r) \vee (q \to r)$ and $(p \wedge q) \to r$ are logically equivalent. [5]

(b) Solve the recurrence relation using generating function method $a_n = 8a_{n-1} + 10^{n-1}$ with initial condition $a_1 = 9$. [5]

4. (a) Find the transitive closure of the given relation using Warshall's algorithm [5]

$R = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$ on $A = \{1,2,3,4\}$.

(b) Let $G = \mathbb{R} - \{0\}$ and $a * b = \frac{ab}{2}, \forall\, a, b \in G$. Show that $(G, *)$ is an abelian group. [5]

5. (a) Find all solutions of the recurrence relation [5]

$$a_n = 5a_{n-1} - 6a_{n-2} + 2^n + 3n.$$

(b) Show that the relation of logical equivalence on the set of all compound propositions is an equivalence relation. What are equivalence classes of F and T. [5]

6. (a) Find the number of elements in $A_1 \cup A_2 \cup A_3$, if there are 100 elements in A_1, 1000 in A_2 and 10,000 in A_3 and [5]

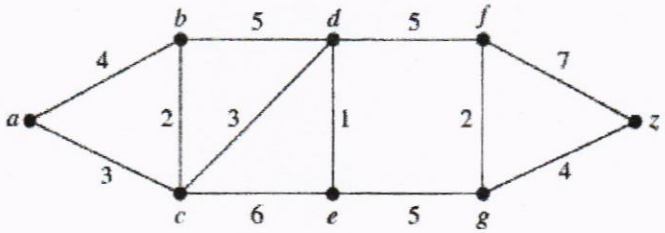
(i) $A_1 \subseteq A_2$ and $A_2 \subseteq A_3$

(ii) the sets are pairwise disjoint

(iii) there are two elements common to each pair of sets and one element in all three sets.

(b) Define degree of a vertex, bipartite graph, incidence and adjacency matrix representation of graph with suitable example. [5]

7. (a) Find the shortest path between a and z in the given weighted graph using Dijistra's algorithim. [5]



(b) Define a lattice. Show that the set $(P(S), \subseteq)$ is a lattice, where $S = \{a, b, c\}$. Draw its Hasse diagram. [5]

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